C32 NMR Spectroscopy – Knowledge Organiser

AQA A-level Chemistry • Set 1: NMR Basics **Paper 2**

#	Question	Answer
1	Standard used in NMR	TMS, $Si(CH_3)_4$
2	Why TMS is used	Single peak; inert; volatile
3	Chemical shift of TMS	0 ppm
4	Symbol and units of chemical shift	δ, ppm
5	Solvent used for NMR	$CDCl_3$
6	Why a deuterated solvent is used	Gives no ^{1}H signal
7	Number of peaks in ^{13}C NMR shows	Number of carbon environments
8	Number of peaks in ^{1}H NMR shows	Number of hydrogen environments
9	Integration ratio shows	Relative number of H in each environment
10	Radiation used in NMR	Radio waves

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AQA A-level Chemistry • Set 2: Splitting **Paper 2**

#	Question	Answer
1	*n* + 1 rule	Peak splits into *n* + 1, where *n* = H on adjacent carbons
2	Singlet	0 H on adjacent carbons
3	Doublet	1 H on adjacent carbons
4	Triplet	2 H on adjacent carbons
5	Quartet	3 H on adjacent carbons
6	Pattern for an ethyl group	Triplet (3H) + quartet (2H)
7	Splitting in ^{13}C NMR spectra	None
8	Pattern for an isolated CH_3 (e.g. next to C=O)	Singlet
9	Where chemical shift values are found	Data booklet
10	Use of NMR in medicine	MRI scanning